Does fiber-aware index calculus ever cross rho? No: the gap grows as #E^1/6

2026-10-11. research/fiber_ic_scaling_20261011. Whole-method cold cost against matched rho, one target per run; negative result. Protocol frozen before the first session; amendment 1 added after reading the second and labelled.

Answer

For each curve, the envelope is the cheapest fiber-closed MITM arm (prime-abscissa:size=B, mitm-fiber closed, coset closure) over B = 4 to 256, divided by rho.negation on the same curve (mean S of 6 runs each).

cofactor	envelope ÷ rho at 18 ⟶ 22 ⟶ 26 ⟶ 30 bits	slope	95 % interval
4	8.4, 8.8 ⟶ 11.0, 11.3 ⟶ 20.6, 17.9 ⟶ 35.1, 37.0	**0.177**	0.145 to 0.209
72 (with amendment 1)	20.1, 13.4 ⟶ 17.7, 21.2 ⟶ 42.5, 37.6 ⟶ 60.4, 63.1	**0.168**	0.115 to 0.221

Both slopes of $\log_2$ of the ratio on $\log_2$ #E match the derived 1/6: a fiber-closed pair table costs about $N^{2/3} h^{-1/3}$ per target against rho's $(N/h)^{1/2}$, so the ratio grows as $(N h)^{1/6}$. The fitted lines reach 1 only at $\log_2$ #E ≈ 0.8 (h = 4) and −5.5 (h = 72). The fiber levers of the conductor-gap rounds lower the IC constant; they cannot change the exponent, and fiber-aware IC falls further behind rho as curves grow.

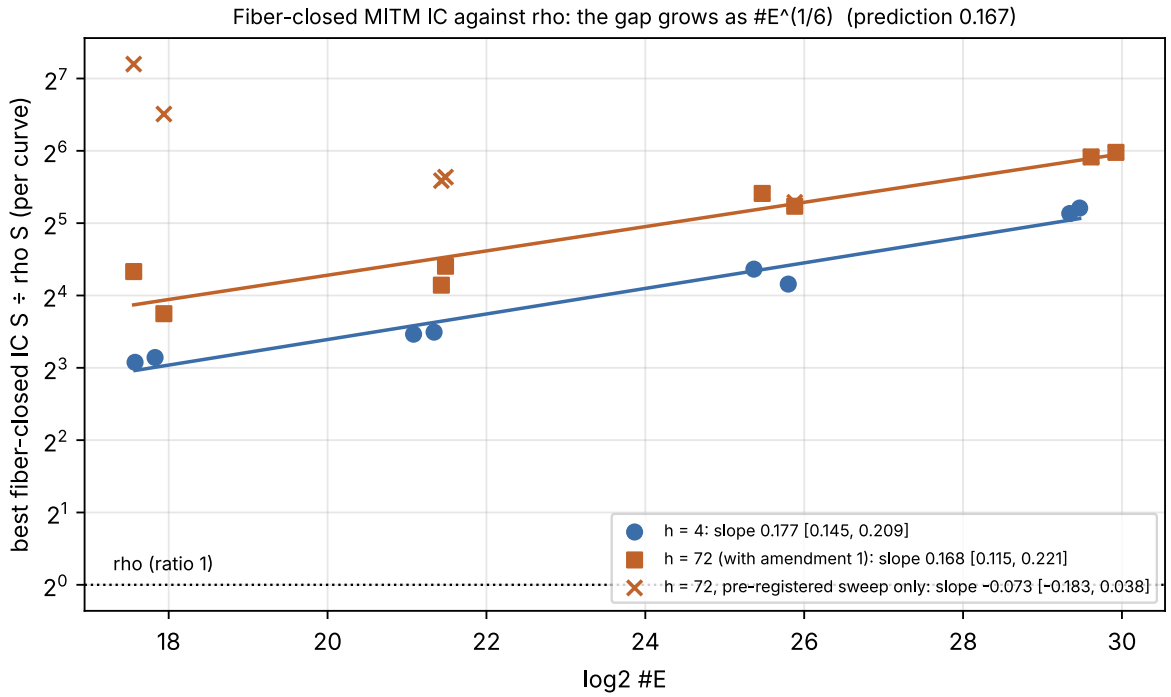


Figure 1: Per-curve envelope $S_{IC} \div S_{rho}$ against $\log_2$ #E, with least-squares fits (tables.json, from examples/fiber_scaling_tables.rs). Crosses: the pre-registered h = 72 sweep alone, whose envelope sat at its floor B = 16 on six of eight curves.

Predictions

- **P-S1 (slope ≈ 1/6, > 0): holds** for both cofactors; for h = 72 only with amendment 1. The pre-registered h = 72 fit alone gives −0.073 (−0.183 to 0.038), an artifact of the unresolved envelope; amendment 1 added B = 4 and 8 on the same curves and targets.
- **P-S2 (h = 72 above h = 4 by ≈ 1.6): holds roughly:** 1.94, 1.74, 2.09, 1.71 at 18, 22, 26, 30 bits.
- **P-S3 (fiber beats blind by ≥ $h^{1/3}$): fails as stated:** blind B = 128 ÷ closed envelope falls to 1.4 (h = 4) and 1.2 to 1.4 (h = 72) at 26 bits.

Sessions

session	records	audit	isolation earned
scaling_h4 (ECBS1h4a5009a8f716)	336	ok, 12 replays	L2 231, L1 105
scaling_h72 (ECBS1hbf4321e2bec8)	336	ok, 12 replays	L2 197, L1 139
scaling_h72_small_b (ECBS1h54835df66b11), amendment 1	144	ok, 12 replays	L2 88, L1 56

Curves: random prime curves with #E = h·r (r prime), two per size at 18, 22, 26, 30 bits for h = 4 and 72, from examples/fiber_scaling_curves.rs. S is ecbench's counted cost over the floor; both sides are lower bounds (IC leaves lookups, square roots, row operations, Legendre symbols and guard probes unpriced; rho its canonicalisations). Run reports: reports/run-20261011T2037Z, T2039Z, T2044Z.

Open

The amortised multi-target regime (one table reused across many targets), where the fiber multiplies the online gain by s for the same table memory, is not measured; the scoreboard keeps it apart from one-target results.